Choose a motion method

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose a motion relationship and justify its assumptions.

SUCCESS CRITERIA - I can...

1. Choose a relationship using the known quantities.
2. Show linked calculations with units and directions.
3. State when constant velocity or acceleration applies.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “constant” and “interval” to explain your model or result. Sentence starters are optional.

constant: not changing during the stated interval

interval: the time or range between two stated values

acceleration: rate of change of velocity

01 Bring back an earlier idea

A cart starts at 3 m and moves right at 5 m/s for 2 s. **Predict** final position. Allowed endpoints are 11 to 15 m; state this interval and one assumption.

Word help: final = at the end of the interval; subscript f

Sentence starter: I predict ... . My model supports this because ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $v_f = v_i + a\Delta t$; $\Delta x = \frac{1}{2}(v_i + v_f)\Delta t$. Constant acceleration; i = initial, f = final. At constant velocity, $\Delta x = v\Delta t$.

Worked reasoning: From rest, a cart accelerates uniformly at 2 m/s² for 3 s. Final velocity=6 m/s; average velocity=(0+6)/2=3 m/s; displacement=3×3=9 m.

Your turn: Show the same 9 m using the area of a triangle on a velocity-time graph.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Choose the relationship

A cart moves steadily at 3 m/s for 4 s. Which relationship fits: steady velocity or changing velocity? **Find** displacement.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

04 Build final velocity

A cart starts at 2 m/s east. It gains 2 m/s each second for 3 s. Complete: gain=2×3=____ m/s; final=2+____=____ m/s east.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Time (s)	Velocity east (m/s)
0	2
1	4
2	6
3	?

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Find the distance while speeding up

Use the cart in 4. Its acceleration is constant. Average velocity=(2+8)÷2=____ m/s. Displacement=average velocity×3 s=____ m.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

06 Choose a method

A cart travels west at 10 m/s and west at 3 m/s 2 s later. Is it speeding up?
Find acceleration size and direction.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

07 Exit check

A cart speeds up uniformly from 0 to 6 m/s east in 5 s. **Find** acceleration, average velocity and displacement. **Show** each step.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Create two different motions that travel 8 m in 4 s: one at steady speed and one speeding up uniformly from rest. **Draw** their velocity-time graphs.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3